Under the Shadow of the Fed: Colombian Macroeconomic Vulnerability to U.S. Rate Shocks

Pablo Alejandro Reyes Granados[1,2] preyesgr@banrep.gov.co
1. Universidad Externado de Colombia
2. Banco de la República

Bayes Plurinacional

Summary. We study the transmission to Colombia of an upward shock in the U.S. 5-year T-bill through an SBVAR with agnostic identification based on sign restrictions, informed by an IS–LM–BP framework calibrated for Colombia. The evidence shows limited monetary autonomy in the face of external shocks and a dominant exchange-rate channel, highlighting the vulnerability of emerging economies to U.S. monetary normalization.

Introduction. Recent increases in U.S. interest rates revive a central question for small open economies: to what extent are external shocks transmitted to domestic financial conditions, prices, and monetary policy? In Colombia, the dominant channel combines *spillovers* from the 5-year T-bill to TES yields, exchange-rate *pass-through* from the TRM to inflation, and a policy rule sensitive to prices and the business cycle. We quantify the *exposure* and *persistence* of these channels with an *SVAR* identified through the *agnostic* methodology of Baumeister–Hamilton (2015). We show how shocks to external rates condition local interest rates, the exchange rate, and inflation, and derive implications for monetary autonomy and risk management.

Methods. *SBVAR and agnostic identification*

Variables. Let $y_t = [\, i_{5y}^{US},\ i_{5y}^{CO},\ \text{TRM},\ \pi,\ i^{CB},\ u\,]'$.

Model: SVAR(p).

$$A_0 y_t = a_0 + \sum_{j=1}^{p} A_j y_{t-j} + C\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \Sigma).$$

Reduced form.

$$y_t = c + \sum_{j=1}^{p} B_j y_{t-j} + u_t, \qquad u_t \sim \mathcal{N}(0, \Omega),$$
$$u_t = A_0^{-1} C \varepsilon_t, \qquad \Omega = A_0^{-1} C \Sigma C' (A_0^{-1})'.$$

Agnostic identification (Baumeister–Hamilton, 2015). We impose *sign restrictions* for $h \in \{0, \ldots, 4\}$ on the shock ε^{US}, derived from an IS–LM–BP framework calibrated for Colombia:

	i_{5y}^{US}	i_{5y}^{CO}	TRM	π	i^{CB}	u
ε^{US}	$+$	$+$	$+$	$+$	$+$	$-$

Note: TRM > 0 denotes depreciation (COP/USD). The signs are treated as *prior information* and implemented through orthogonal rotations that satisfy those directions.

Priors and computation. Minnesota/dummy priors for (B, Ω); weak priors for (A_0, Σ) with a penalty for violating sign restrictions.
MCMC: Gibbs for (B, Ω) and adaptive Metropolis–Hastings for (A_0, C). IRF/FEVD with credible bands (68/80 %) and shocks normalized to 1 sd.

Results. *Summary of key findings (IRF and FEVD).*

- **Immediate transmission:** an $\uparrow$ in i_{5y}^{US} raises i_{5y}^{CO} and i^{CB}; the **TRM** depreciates; π exhibits an early *hump-shaped* profile; u falls.
- **Persistence:** the effects on interest rates dissipate within 18–24 months; the response of i^{CB} peaks around $h \approx 6$–8 (with h in months).
- **Uncertainty:** credible bands (68/80 %) are tight at short horizons and become wider over longer horizons.
- **FEVD:** the external shock explains a *substantial* share of the variance in i_{5y}^{CO} and i^{CB}; in the TRM, its own shock dominates, but the external one gains importance with the horizon.
- **Robustness (LP):** *Local Projections* responses replicate the signs and temporal ordering of the SBVAR IRFs.

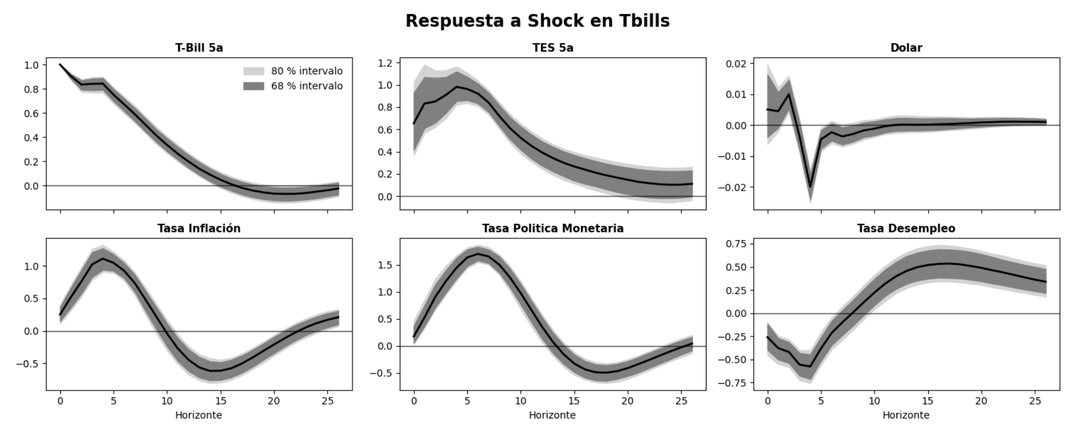

Figure 1: Impulse responses (IRF) to an external shock in i_{5y}^{US}. Credible bands 68/80 %.

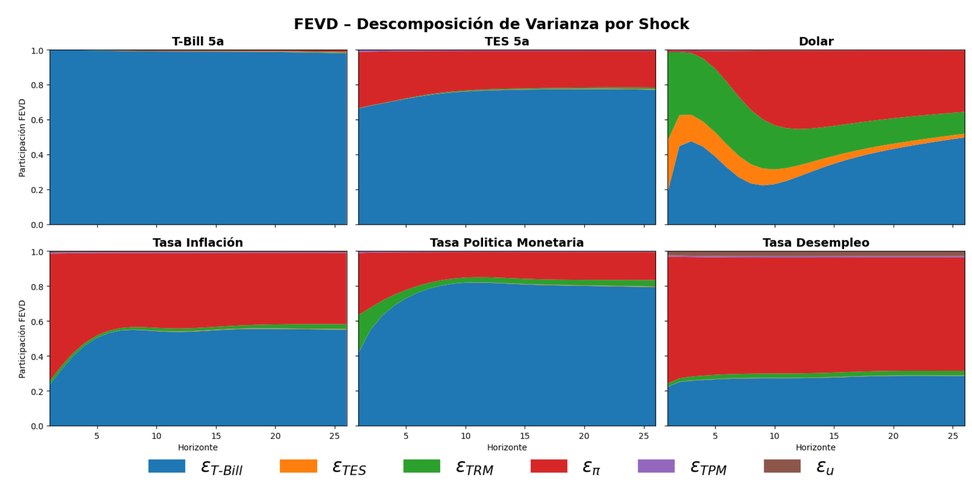

Figure 2: Forecast error variance decomposition (FEVD) by shock.

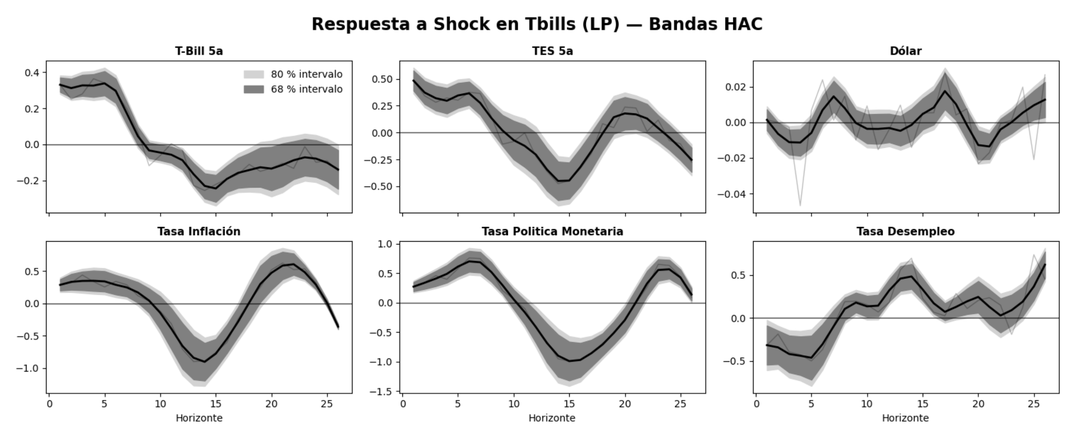

Figure 3: Impulse responses estimated using Local Projections (LP) compared with the SBVAR IRF. Credible bands 68/80 %.

Conclusion. Overall, the evidence suggests that increases in i_{5y}^{US} significantly shape Colombia's macro-financial dynamics: they raise i_{5y}^{CO}, put depreciation pressure on the TRM, accelerate π, and force a reaction in i^{CB} that, although effective in the short run, reveals limited monetary autonomy. This points to the need to strengthen exchange-rate hedging, liquidity management, and forward guidance to cushion external shocks.

References

[1] Baumeister C, Hamilton JD. Sign Restrictions, Structural Vector Autoregressions, and Useful Prior Information. Econometrica. 2015;83(2):689–733.

[2] Jordà Ò. Estimation and Inference of Impulse Responses by Local Projections. American Economic Review. 2005;95(1):161–182.

[3] Jacobo J. Una introducción a los métodos de máxima entropía y de inferencia bayesiana en econometría. Bogotá, Colombia: Universidad Externado de Colombia; 2025.